ARCHIT AGRAWAL

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PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Own a full-stack TypeScript platform end to end: a 14-page Nuxt 4 frontend over a 10-module NestJS API sharing one typed contract, with 4,845 tests across 347k lines in 12 workspace packages.
- Shipped an AI chart builder: readers ask in plain English, get a Vega-Lite chart, then edit it with sort, stack, top-N, filters and undo/redo; every chart also ships an accessible data table and alt text.
- Built EdSpace in two days for the EdPlus hackathon: a live office map with room booking, desk claiming, walking routes and 3D desks, shown to leadership up to the CEO and going live for 500+ people.
- Accelerated publishing of a 51,280-respondent survey from 19 minutes to 109 seconds; an independent raw-data audit matched all evaluated figures.
- Delivered interactive chart sorting, filtering, top-N and undo in Nuxt; in-process DuckDB kept measured query latency flat as input rows grew 200x.
- Cut end-to-end AWS pipeline runtime from 500 to 85 seconds by removing managed-agent startup overhead and invoking the underlying tools directly.
- Secured the analytics proof of concept with entitlement-gated row-level access, Bedrock Guardrails deployed through AWS CDK and enforced per-stage spend limits.

MyStage Music Inc. **03/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Delivered sub-50 ms venue search through FastAPI and Algolia, synchronizing Firestore updates into the search index in real time.
- Reduced production service dependencies 64% by consolidating a 14-Cloud-Function backend into a checkpointed Python/LangGraph workflow on GCP Cloud Run.
- Expanded Playwright ingestion to 70,000+ records daily from 650+ venues, using proxy rotation, backoff and transactional locking to handle unreliable sources.
- Covered routing, transactional claims, retries and checkpoint resumes with 200+ pytest unit and integration tests; traced workflow execution through Logfire.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Built a multi-tenant course assistant from React UI to retrieval services, used by 1,000+ faculty creating courses for 60,000+ students.
- Increased assessment-tool engagement 35% and reduced bounce 20% by building responsive React and Material UI workflows for bulk import, question reuse and revision comparison.
- Built REST APIs and SQL-backed admin tools for ASU Online quiz platforms and question banks, with schema-level invariants protecting assessment integrity.
- Enabled configuration-based college onboarding through tenant-scoped retrieval, keeping each college's documents within its own search boundary.
- Cut transcript-analysis time from four hours to 15 minutes with a Neo4j knowledge graph and schema-validated, LLM-generated Cypher queries.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Delivered 20+ production REST APIs using .NET 8, Dapper and MediatR, with JWT authentication and AWS SQS for queued work.
- Cut API latency from 198 to 20 ms with Redis caching and SQL rewrites, verified under simulated peak load with Locust.
- Built an idempotent ETL pipeline for 10,000+ researcher profiles daily, using PostgreSQL natural-key upserts to prevent duplicates after partial failures.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Built React and Redux apps with normalized client state and typed REST integration, raising measured UX scores 40%.
- Migrated a C#/.NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and resource footprint 35%.
- Reduced deployment time 70% and production incidents 40% through containerized CI/CD and SonarQube quality gates.
- Improved student-listing performance 30% with pagination and S3 asset delivery; reduced class-detail latency 10% through SQL query rewrites.

EAT.FIT **09/2021 - 12/2021**
Product Intern *Bengaluru, India*

- Cut driver-tracking server cost 45% with distance-based polling and client-side interpolation that kept map movement smooth.
- Built a WebSocket and Express notification service with backpressure-aware fan-out and duplicate-safe redelivery; reported CSAT rose 28% and support volume fell 35%.

PROJECTS

Clash Royale Analytics **2025**

- Built a serverless analytics platform on Cloudflare Workers and D1 with Drizzle ORM, fronted by Next.js 16 and React 19 on Vercel, with cross-filtered D3 visualizations.
- Shipped accessibility with skip navigation, focus traps, reduced-motion support and 44-pixel touch targets.

PrismSplit **2025 - Present**

- Built a TypeScript/React Native expense app with transactional PostgreSQL billing and settlement, integer-cent accounting and offline write persistence.
- Added idempotency keys, real-time Supabase subscriptions and versioned persistence to handle retries and cached data across app upgrades.
- Wrote Jest unit and interaction tests, Maestro end-to-end flows and Android release-build checks.

E-Commerce Platform **09/2024 - 12/2024**

- Built an e-commerce platform with a Django REST API, React frontend, Stripe payments, PostgreSQL with Redis caching, Docker and GitHub Actions CI.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C#, HTML, CSS

Frontend: React, Next.js, Vue/Nuxt, Angular, React Native, Redux, Zustand, TanStack Query, Tailwind CSS, Material UI, D3

Backend & APIs: Node.js, NestJS, Express, FastAPI, Django, Flask, .NET 8, REST APIs, WebSockets, JWT

Databases & Search: PostgreSQL, MySQL, Redis, DynamoDB, Firestore, Supabase, DuckDB, Algolia, Neo4j, pgvector

Cloud & DevOps: AWS (Lambda, API Gateway, S3, SQS, Step Functions, CDK), GCP Cloud Run, Vercel, Cloudflare Workers, Docker, Kubernetes, GitHub Actions

Testing & Observability: Jest, Vitest, React Testing Library, Maestro, pytest, Locust, SonarQube, Logfire, OpenTelemetry

AI Applications: AWS Bedrock, OpenAI, LangGraph, RAG, LLM evals, MCP

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*